

p0408r7 - Efficient Access to `basic_stringbuf`'s Buffer

Including wording from p0407 Allocator-aware `basic_stringbuf`

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1 Motivation

Streams have been the oldest part of the C++ standard library and their specification doesn't take into account many things introduced since C++11. One of the oversights is that there is no non-copying access to the internal buffer of a `basic_stringbuf` which makes at least the obtaining of the output results from an `ostreamstream` inefficient, because a copy is always made. I personally speculate that this was also the reason why `basic_strbuf` took so long to get deprecated with its `char *` access.

With move semantics and `basic_string_view` there is no longer a reason to keep this pessimisation alive on `basic_stringbuf`.

I also believe we should remove `basic_strbuf` from the standard's appendix [depr.str.strstreams]. This is proposed in p0448, that completes the replacement of that deprecated feature.

2 Introduction

This paper proposes to adjust the API of `basic_stringbuf` and the corresponding stream class templates to allow accessing the underlying string more efficiently.

C++17 and library TS have `basic_string_view` allowing an efficient read-only access to a contiguous sequence of characters which I believe `basic_stringbuf` has to guarantee about its internal buffer, even if it is not implemented using `basic_string` obtaining a `basic_string_view` on the internal buffer should work sidestepping the copy overhead of calling `str()`.

On the other hand, there is no means to construct a `basic_string` and move from it into a `basic_stringbuf` via a constructor or a move-enabled overload of `str(basic_string &&)`.

2.1 History

2.1.1 Changes from r6

The feedback by LWG in Cologne 2019 was incorporated.

- bump paper revision and base it on current working draft (Daniel Krügler checked that all existing context wording is still OK.)
- specify the moved from state to be "`str().empty()` is `true`" instead of comparing with an empty string literal.
- make initialization sequencing more clear by replacing ". Calls..." with ", then calls..."
- reformulate constraint on constraint constructor (taking a string with a different allocator) to be in line with current wording guidelines (sentence instead of code). (also for `str(SAlloc)` overload).
- simplified effects clause of constructors moving from string.
- make conditional `noexcept` conform with synopsis for `swap`, but non-member `swap` gets it in synopsis without see below, because it is short enough.
- remove obsolete wording in constructor definitions ("Constructs..." to "Initilizes")
- replace many "Returns:" with "Effects: Equivalent to: " to obtain requirements in many places. This allows to remove some Constraints on `SAlloc` (if Marshall believes so - Tomasz said to remove the `str()` constraints on `SAlloc` is OK).
- replace many "Calls " with "Equivalent to:" in Effects clauses
- simplifies wording for `str()` member functions by relating to `view()` (except for rvalue-ref qualified one)
- make `view()` member specification simpler and more correct, by using "Let `sv` be `basic_string_view<....>`" and referring to `sv` instead of the incomplete type right now.
- minor cosmetic adjustments wrt spacing
- make stream swap operations more clear and implicitly add preconditions through specifying them with effects equivalent to (as suggested by Tomasz Kaminski).
- consistently make `view()` `noexcept`
- add `get_allocator()` to `str()` calling `view`.

2.1.2 Changes from r5

There was a review in my absence (again) in San Diego, November 2018. I'll try my best to incorporate the feedback here.

- rebase on n4791.
- undo premature application of p1163 (explicit -> non-explicit multi-parameter ctors by additional overloads) (ARGHHH, but I now think I follow Titus argumentation that it might be a bad idea).
- see table 1. LEWG might need to reconsider the combination of p0407/p0408 to agree on sane ctor overloads. LWG and Ville gave feedback on different ctor overloads. New Design:

separate SFINAE'd overloads for "foreign allocator" string arguments.

- clean up `str()` member function overloads. This was in the overlap of p0407 and p0408 and not seen by LEWG in that way (sorry!). Split getter to two, one taking an allocator for the new string. Setter `str(string const&)` member function remains a template on the string's Allocator.
- adjust the *italic* explanations accordingly to the changes.
- LWG question: Do the constructors taking a `SAlloc` template parameter restrict it to *Cpp17Allocator* requirements? It is implicit via `basic_string`.
- Fixed a specification bug in move construction allowing keeping the original wording of move-assign. `rhs` must be "synced" first, by relying on initializing `buf` from `std::move(rhs).str()` instead from `std::move(rhs.buf)` directly.
- merge getters `str()` specification of `high_mark` into a single specification for simplification and consistency.
- split copying setters `str(basic_string const &)` into the previously existing one and the one taking a `basic_string` with a different allocator, like with the constructors to reduce ABI problems.
- drive by editorial fix to mention already existing Allocator template parameter in stream classes, i.e., `basic_istream<charT, traits, Allocator>` where mentioned in descriptions
- drive-by fix to postcondition of `basic_stringbuf` move constructor to also refer to `getloc()`.

2.1.3 Changes from r4

Incorporate suggestions from LWG review in Batavia, August 2018. This was the first time the combined proposal was reviewed.

- Adjust specification sections to new naming schema introduced at the Rapperswil Meeting 2018 for C++20. (Requires->(Mandates (compile-time), Expects(contract)), Remarks->Constraints, Postconditions->Ensures).
- change the overloads of constructors with default arguments to only have the single argument version explicit according to p1163.
- introduce an exposition-only member function `init_buf_ptrs()` in `basic_stringbuf` to set the streambuf pointers. In the standard version, that was part of the `str(string)` member function and now is needed in more than one place. Add a note there about internally violating invariants of `buf`. Explain the exposition only members in the front matter of the class.
- reduce clutter, since bit operations are possible in enum `ios_base::openmode` parameters.
- Provide a note that allocator properties are propagated along the properties of the `basic_string` member `buf` in the front matter of the class. I hope this is sufficient to address the issue from Batavia about what happens with the allocators. Also all other allocator relevant comments should be addressed through that delegation to `basic_string`'s properties.
- I provided the following definition of `swap` for `basic_stringbuf` adopted from `basic_string`. Note, the base class swap does not give a noexcept guarantee. I provided that:

Table 1: Overview of stringbuf/stringstream constructors

string	which	Allocator	ctor	comment
			default	exists
	yes		explicit	exists
copy	opt		explicit	exists
	<u>yes</u>	<u>yes</u>		407 for stateful allocs
<u>move</u>	<u>opt</u>		<u>explicit</u>	408, combined again
<u>other-copy</u> ¹		<u>yes</u>		407 other kind of strings ²
<u>other-copy</u> ¹	<u>yes</u>	<u>yes</u>		407 other kind of strings ²
<u>other-copy</u> ¹	<u>opt</u>		<u>explicit</u>	407 above with default alloc ³
		<u>yes</u>	<u>explicit</u>	407 for stateful allocators
<u>move</u>	<u>yes</u>	<u>yes</u>		408r5 - useless, copies anyway

1 other-copy means has a different Allocator template argument

2 allow if same or different allocator for string, because allocator is given, see copy-ctor string with allocator parameter.

3 LEWG new design question: requires string Allocator different from stringbuf Allocator (new), otherwise existing ctor is changed ABI (and may be CTAD) breakage.

* in addition a move ctor is defined taking an additional Allocator argument like with basic_string

** Allocator should always be the last Parameter (is that really always the case?).

```
void swap(basic_stringbuf& s)
    noexcept(allocator_traits<Allocator>::propagate_on_container_swap::value ||
             allocator_traits<Allocator>::is_always_equal::value);
```

- `basic_stringbuf` move constructor now guarantees that `rhs` is empty, as if `std::move(rhs).str()` was called. This seems the easiest way to guarantee its get and put area are re-initialized accordingly. While technically not required, it makes handling moved-from streambufs (which are rare) consistent with calling the rvalue-ref-qualified `str()` member function. Please note, we do not give such a guarantee to the moved-from state of the stream objects, since they get their buffer pointer stolen and thus are completely unrelated to a streambuf after been moved-from. Only moving out the underlying string keeps the stream in working condition.

2.1.4 Changes from r3

To make the job of reviewing and integrating my stringstream adjustments easier, I incorporate the changes proposed in p0407r2 (allocator-aware `basic_stringbuf`), since both papers have been forwarded by LEWG to LWG.

- Added full set of reasonable overloads to the constructors with and without allocator (`basic_string&&` does not get an allocator constructor template argument to allow efficient construction from `charT*` literals).

2.1.5 Changes from r2

Discussed in Albuquerque, where LEWG was in favor to forward it to LWG for IS with the following change.

- reestablish rvalue-ref qualified `str()` instead of the previously suggested `pilfer()`.
- address LWG only in document header.

2.1.6 Changes from r1

Discussed in LEWG Issaquah. Answering some questions and raising more. Reflected in this paper.

- reflected new section numbers from the std. now relative to the current working draft.
- implementation is now working with gcc 7. (not relevant for this paper)

2.1.7 Changes from r0

- Added more context to synopsis sections to see all overloads (Thanks Alisdair).
- rename `str_view()` to just `view()`. There was discussion on including an explicit conversion operator as well, but I didn't add it yet (my implementation has it).
- renamed r-value-ref qualified `str()` to `pilfer()` and removed the reference qualification from it and remaining `str()` member.
- Added allocator parameter for the `basic_string` parameter/result to member functions (see p0407 for allocator support for stringstreams in general)

3 Acknowledgements

- Daniel Krüger encouraged me to pursue this track.
- Alisdair Meredith for telling me to include context in the synopsis showing all overloads. That is the only change in this version, no semantic changes!
- Jonathan Wakely to show me the `#undef _GLIBCXX_EXTERN_TEMPLATE`

4 Impact on the Standard

This is an extension to the API of `basic_stringbuf`, `basic_stringstream`, `basic_istringstream`, and `basic_ostringstream` class templates.

~~This paper addresses both Library Fundamentals TS 3 and C++ Next (2020?).~~ When added to the standard draft with p0448 (spanstream), section [depr.str.strstreams] should be removed.

5 Design Decisions

After experimentation I decided that substituting the `(basic_string<charT, traits, Allocator const &)` constructors in favor of passing a `basic_string_view` would lead to ambiguities with the new move-from-string constructors.

5.1 Hint to implementers

In both libc++ and libstdc++ I needed to make `basic_stringbuf` a friend of `basic_string` to allow efficient growth of the buffer beyond the current string length (breaking an invariant) until it is retrieved using one of the `str()` member functions. Other implementations might use a different strategy of caring for the buffer space that should be efficiently be adopted by the returned string object, thus requiring either special `basic_string` constructors or access to its internals as well.

5.2 Open Issues to be discussed by LWG

Note: this list includes the discussion of p0407 features.

- *Does it make sense to add `noexcept` specifications for `move()` and `swap()` members, since the base classes and other streams do not. At least it does not make sense so for stream objects, since the base classes do not specify that.*
- ~~The `basic_string` constructors that move from the string get a default template argument for `SAlloc` in the hope that allows initialization from a character string literal. Need confirmation that this trick works and selects the better constructor for temporary conversion without ambiguity, because for the copying (const-ref) overload the allocator of the string needs to be deduced. This should lead to the effect of optimizing existing usages.~~

5.3 Open Issues discussed by LEWG in Albuquerque

- Should `pilfer()` be rvalue-ref qualified to denote the "destruction" of the underlying buffer? LEWG in Issaquah didn't think so, but I'd like to ask again. LEWG small group in Albuquerque in favor of rvalue-ref qualification. [Re-establish `str\(\)&&`, drop `pilfer`](#)

5.4 Open Issues discussed by LEWG in Issaquah and Albuquerque

- Is the name of the `str_view()` member function ok? No. Renamed to `view()`
- Should the `str()&&` overload be provided for move-out? ~~No. give it another name (`pilfer`) and remove rvalue-ref qualification (Issaquah).~~ [Re-establish `str\(\)&&`, drop `pilfer`](#)
- Should `str()&&` empty the character sequence or leave it in an unspecified but valid state? Empty it, and specify.
- Provide guidance on validity lifetime of of the obtained `string_view` object.

5.5 Open Issues to be discussed by LEWG/LWG (in Kona?)

- LEWG: Please look at constructor overloads (see Table 1) and `str()` overloads that came from the mix of p0407 with p0408.
- Both: Constructor overloads taking a string with a different allocator, esp. SFINAE. Is that OK?
- LWG: Is `!is_same_v<SAlloc, Allocator>` the correct SFINAE predicate for foreign allocator overloads?
- LEWG: `noexcept` for `view()` member function of `stringbuf` (note streams have a precondition on this call and can not be `noexcept`)(suggested by LWG).

- LWG: Does an Allocator template parameter that is mapped to `basic_string`'s Allocator template parameter need to conform to *Cpp17Allocator* requirements or is that given implicit by its usage?
- LWG: recheck wording.

6 Technical Specifications

The following is relative to n4820.

Remove section on `char*` streams [depr.str.strstreams] and all its subsections from appendix D.

6.1 28.8.2 Adjust synopsis of `basic_stringbuf` [stringbuf]

Add a new constructor overload.

Note that p0407 provides allocator support for `basic_stringbuf`, since now both papers have been forwarded to LWG, the changes proposed in p0407 are integrated here for ease of review and integration. The explanations of those changes are added in italics here. from r6 on some changes that need to be revisited by LEWG are made, since the overlap of the two papers' functionality.

Change each of the non-moving, non-deleted constructors to add a const-ref `Allocator` parameter as last parameter with a default constructed `Allocator` as default argument. Add an overload for the move constructor adding an `Allocator` parameter like with `basic_string`. Add an exposition-only member variable `buf` to allow referring to it for specifying allocator behaviour. May be: Add `noexcept` specification, depending on allocator behavior, like with `basic_string`?

This section also adopts the changes of p1163 by only marking the single argument constructors explicit and provide non-explicit overloads for zero, two or more argument versions. That paper p1163 was tentatively accepted in Batavia, August 2018.

```
// [stringbuf.cons], constructors
basic_stringbuf() : basic_stringbuf(ios_base::in | ios_base::out) {}
explicit basic_stringbuf(ios_base::openmode which);
explicit basic_stringbuf(
    const basic_string<charT, traits, Allocator>& str,
    ios_base::openmode which = ios_base::in | ios_base::out);

explicit basic_stringbuf(const Allocator& a)
    : basic_stringbuf(ios_base::in | ios_base::out, a) { }

basic_stringbuf(ios_base::openmode which, const Allocator& a);
explicit basic_stringbuf(
    basic_string<charT, traits, Allocator>&& s,
    ios_base::openmode which = ios_base::in | ios_base::out);
template<class SAlloc>
basic_stringbuf(
    const basic_string<charT, traits, SAlloc>& s,
    const Allocator& a)
    : basic_stringbuf(s, ios_base::in | ios_base::out, a) { }
template<class SAlloc>
basic_stringbuf(
```

```

    const basic_string<charT, traits, SAlloc>& str,
    ios_base::openmode which,
    const Allocator& a);
template<class SAlloc>
basic_stringbuf(
    const basic_string<charT, traits, SAlloc>& str,
    const Allocator& a) : basic_stringbuf(str, ios_base::in | ios_base::out, a) {}
template<class SAlloc>
explicit basic_stringbuf(
    const basic_string<charT, traits, SAlloc>& str,
    ios_base::openmode which = ios_base::in | ios_base::out);
basic_stringbuf(const basic_stringbuf& rhs) = delete;
basic_stringbuf(basic_stringbuf&& rhs);
basic_stringbuf(basic_stringbuf&& rhs, const Allocator& a);

// [stringbuf.assign], assign and swap
basic_stringbuf& operator=(const basic_stringbuf& rhs) = delete;
basic_stringbuf& operator=(basic_stringbuf&& rhs);
void swap(basic_stringbuf& rhs) noexcept(see below);

```

The following list summarizes the edits:

- Add an rvalue-ref overload of `str()` that obtains the underlying string via moving from `buf`.
- Add a `str(Allocator)` overload template member function to take an `Allocator` for the returned string and add a reference qualification the existing `str()` overload. NEW: was intermingled with existing `str()` member, now separate.
- Add the `view()` member function obtaining a `string_view` to the underlying internal buffer. NEW: make that `noexcept`.
- Add a setter `str()` overload as a template member function copying into the string buffer to take an allocator template parameter that differs from the buffer's own `Allocator`
- Add a `str(string&&)` overload that moves from its string rvalue-reference argument into the internal buffer.
- Provide an exposition-only member function `init_buf_ptrs()` to ensure streambuf pointers are initialized correctly by all `buf` setting operations.

```

// [stringbuf.members], getters and setters:
basic_string<charT, traits, Allocator> str() const &;

template<class SAlloc>
basic_string<charT, traits, SAlloc> str(const SAlloc& sa) const;
basic_string<charT, traits, Allocator> str() &&;
basic_string_view<charT, traits> view() const noexcept;

void str(const basic_string<charT, traits, Allocator>& s);

template<class SAlloc>
void str(const basic_string<charT, traits, SAlloc>& s);
void str(basic_string<charT, traits, Allocator>&& s);

```


Add the following declaration to the public section of synopsis of the class template `basic_stringbuf`:

```
allocator_type get_allocator() const noexcept;
```

Add the following exposition only member to the private section of synopsis of the class template `basic_stringbuf`. This allows to delegate all details of allocator-related behaviour on what `basic_string` is doing, simplifying this specification a lot.

```
private:
    ios_base::openmode mode; // exposition only
    basic_string<charT, traits, Allocator> buf; // exposition only
    void init_buf_ptrs(); // exposition only
```

Add a conditional `noexcept` specification to swap with see below:

```
template <class charT, class traits, class Allocator>
    void swap(basic_stringbuf<charT, traits, Allocator>& x,
              basic_stringbuf<charT, traits, Allocator>& y) noexcept(noexcept(x.swap(y)));
```

Adjust p2 of the section to include the additional exposition only members and add a note on the allocator properties of `basic_stringbuf`.

- 1 The class `basic_stringbuf` is derived from `basic_streambuf` to associate possibly the input sequence and possibly the output sequence with a sequence of arbitrary *characters*. The sequence can be initialized from, or made available as, an object of class `basic_string`.
- 2 For the sake of exposition, the maintained data and internal pointer initialization is presented here as:
 - (2.1) — `ios_base::openmode mode`, has in set if the input sequence can be read, and out set if the output sequence can be written.
 - (2.2) — `basic_string<charT, traits, Allocator> buf` contains the underlying character sequence.
 - (2.3) — `init_buf_ptrs()` sets the base class' get area (`[streambuf.get.area]`) and put area (`[streambuf.put.area]`) pointers after initializing, moving from, or assigning to `buf` accordingly.

6.1.1 28.8.2.1 `basic_stringbuf` constructors [stringbuf.cons]

Adjust the constructor specifications taking the additional `Allocator` parameter and an overload for the move-constructor taking an `Allocator`. Make the constructors' wording that actually construct a `buf` consistent.

```
explicit basic_stringbuf(ios_base::openmode which);
```

- 1 *Effects:* ~~Constructs an object of class `basic_stringbuf`, initializing~~ Initializes the base class with `basic_streambuf()` (`[streambuf.cons]`), and ~~initializing~~ mode with `which`. It is implementation-defined whether the sequence pointers (`eback()`, `gptr()`, `egptr()`, `pbase()`, `pptr()`, `epptr()`) are initialized to null pointers.
- 2 *Ensures:* `str().empty()` is true ~~== ""~~.

```
explicit basic_stringbuf(
    const basic_string<charT, traits, Allocator>& s,
    ios_base::openmode which = ios_base::in | ios_base::out);
```

- 3 *Effects:* ~~Constructs an object of class `basic_stringbuf`, initializing~~Initializes the base class with `basic_streambuf()` (`[streambuf.cons]`), ~~and initializing~~ mode with which, and buf with s, then calls `init_buf_ptrs()`. ~~Then calls `str(s)`.~~

```
basic_stringbuf(
    ios_base::openmode which,
    const Allocator &a);
```

- 4 *Effects:* Initializes the base class with `basic_streambuf()` (`[streambuf.cons]`), mode with which, and buf with a, then calls `init_buf_ptrs()`.
- 5 *Ensures:* `str().empty()` is true.

```
explicit basic_stringbuf(
    basic_string<charT, traits, Allocator>&& s,
    ios_base::openmode which = ios_base::in | ios_base::out);
```

- 6 *Effects:* ~~Constructs an object of class `basic_stringbuf`, initializing~~Initializes the base class with `basic_streambuf()` (`[streambuf.cons]`), ~~initializing~~ mode with which, and buf with `std::move(s)`, then calls `init_buf_ptrs()`.

```
template<class SAlloc>
basic_stringbuf(
    basic_string<charT, traits, SAlloc>& s,
    ios_base::openmode which,
    const Allocator &a);
```

- 7 *Effects:* Initializes the base class with `basic_streambuf()` (`[streambuf.cons]`), mode with which, and buf with `{s,a}`, then calls `init_buf_ptrs()`.

```
template<class SAlloc>
explicit basic_stringbuf(
    const basic_string<charT, traits, SAlloc>& s,
    ios_base::openmode which = ios_base::in | ios_base::out);
```

- 8 *Constraints:* `is_same_v<SAlloc, Allocator>` is false.
- 9 *Effects:* Initializes the base class with `basic_streambuf()` (`[streambuf.cons]`), mode with which, and buf with s, then calls `init_buf_ptrs()`.

Add the additional move constructor taking an allocator and adjust the description accordingly:

```
basic_stringbuf(basic_stringbuf&& rhs);
basic_stringbuf(basic_stringbuf&& rhs, const Allocator& a);
```

Note to LWG reviewers: using `std::move(rhs).str()` ensures `rhs.buf` is in a consistent state before the move happens. Before the spec was wrong, because `rhs.buf` might have been shorter than the actual written characters. Also a drive by (IMHO editorial fix) better spelling out what happens since we now have the exposition only members.

- 10 *Effects:* ~~Move constructs from the rvalue `rhs`.~~ Copy constructs the base class from `rhs` and initializes mode with `rhs.mode`. In the first form buf is initialized from `std::move(rhs).str()`. In the second form buf is initialized from `{std::move(rhs).str(), a}`. It is implementation-defined whether the sequence pointers in `*this` (`eback()`, `gptr()`, `egptr()`, `pbase()`, `pptr()`,

`epptr()`) obtain the values which `rhs` had. ~~Whether they do or not, `*this` and `rhs` reference separate buffers (if any at all) after the construction. The `openmode`, `locale` and any other state of `rhs` is also copied.~~

11 *Ensures:* Let `rhs_p` refer to the state of `rhs` just prior to this construction and let `rhs_a` refer to the state of `rhs` just after this construction.

```
(11.1) — str() == rhs_p.str()
(11.2) — gptr() - eback() == rhs_p.gptr() - rhs_p.eback()
(11.3) — egptr() - eback() == rhs_p.egptr() - rhs_p.eback()
(11.4) — pptr() - pbase() == rhs_p.pptr() - rhs_p.pbase()
(11.5) — epptr() - pbase() == rhs_p.epptr() - rhs_p.pbase()
(11.6) — if (eback()) eback() != rhs_a.eback()
(11.7) — if (gptr()) gptr() != rhs_a.gptr()
(11.8) — if (egptr()) egptr() != rhs_a.egptr()
(11.9) — if (pbase()) pbase() != rhs_a.pbase()
(11.10) — if (pptr()) pptr() != rhs_a.pptr()
(11.11) — if (epptr()) epptr() != rhs_a.epptr()
(11.12) — getloc() == rhs_p.getloc()
(11.13) — rhs is empty but usable, as if std::move(rhs).str() was called.
```

6.2 28.8.2.2 Assign and swap [stringbuf.assign]

Most of this section is included to allow for simpler adding of conditional `noexcept`.

```
basic_stringbuf& operator=(basic_stringbuf&& rhs);
```

1 *Effects:* After that move assignment `*this` has the observable state it would have had if it had been move constructed from `rhs` (see [stringbuf.cons]).

2 *Returns:* `*this`.

```
void swap(basic_stringbuf& rhs) noexcept(see below);
```

3 *Expects:* `allocator_traits<Allocator>::propagate_on_container_swap::value` is true or `get_allocator() == s.get_allocator()` is true.

4 *Effects:* Exchanges the state of `*this` and `rhs`.

5 *Remarks:* The expression inside `noexcept` is equivalent to:
`allocator_traits<Allocator>::propagate_on_container_swap::value ||`
`allocator_traits<Allocator>::is_always_equal::value.`

```
template <class charT, class traits, class Allocator>
void swap(basic_stringbuf<charT, traits, Allocator>& x,
          basic_stringbuf<charT, traits, Allocator>& y) noexcept(noexcept(x.swap(y)));
```

6 *Effects:* Equivalent to: `x.swap(y)`.

6.2.1 28.8.2.3 Member functions [stringbuf.members]

Provide a section introducing paragraph explaining the high-water-mark. The wording is taken directly from n4791 [stringbuf.members] p.1 with some grammar adjustment to adjust to the fact that we now have multiple setters. Introduce the exposition only private member functions `init_buf_ptrs()` to provide the correct initialization of streambuf pointer members and adjust the `str(s)` member functions with parameters accordingly.

- ¹ The member functions getting the underlying character sequence all refer to a `high_mark` value, where `high_mark` represents the position one past the highest initialized character in the buffer. Characters can be initialized by writing to the stream, by constructing the `basic_stringbuf` passing a `basic_string` argument, or by calling one of the `str()` member functions passing a `basic_string` as an argument. In the latter case, all characters initialized prior to the call are now considered uninitialized (except for those characters re-initialized by the new `basic_string`).

```
void init_buf_ptrs(); // exposition only
```

- ² *Effects:* Initializes the input and output sequences from `buf` according to mode.

- ³ *Ensures:*

- (3.1) — If `ios_base::out` is set in mode, `pbase()` points to `buf.front()` and `epptr() >= pbase() + buf.size()` is true;
- (3.1.1) — in addition, if `ios_base::ate` is set in mode, `pptr() == pbase() + buf.size()` is true,
- (3.1.2) — otherwise `pptr() == pbase()` is true.
- (3.2) — If `ios_base::in` is set in mode, `eback()` points to `buf.front()`, and `(gptr() == eback() && egptr() == eback() + buf.size())` is true.
- ⁴ [Note: For efficiency reasons stream buffer operations might violate invariants of `buf` while it is held encapsulated in the `basic_stringbuf`, i.e., by writing to characters in the range `[buf.data()+buf.size(), buf.data()+buf.capacity())`. All operations retrieving a `basic_string` from `buf` ensure that the `basic_string` invariants hold on the returned value. — end note]

Add the definition of the `get_allocator` function:

```
allocator_type get_allocator() const noexcept;
```

- ⁵ *Returns:* `buf.get_allocator()`.

Add a getter overload taking an allocator parameter for the copied from string to allow having a different allocator than the underlying stream and add a ref-qualifier to the existing getter overload to avoid ambiguities with the rvalue-ref qualified overload. Add a getter overload that is rref qualified and mention it. Simplify wording by delegating to the new `view()` member.

```
basic_string<charT, traits, Allocator> str() const &;
```

- ⁶ ~~*Returns:* A `basic_string` object whose content is equal to the `basic_stringbuf` underlying~~

character sequence. If the `basic_stringbuf` was created only in input mode, the resultant `basic_string` contains the character sequence in the range `[eback(), egptr())`. If the `basic_stringbuf` was created with `which & ios_base::out` being nonzero then the resultant `basic_string` contains the character sequence in the range `[pbase(), high_mark)`, where `high_mark` represents the position one past the highest initialized character in the buffer. Characters can be initialized by writing to the stream, by constructing the `basic_stringbuf` with a `basic_string`, or by calling the `str(basic_string)` member function. In the case of calling the `str(basic_string)` member function, all characters initialized prior to the call are now considered uninitialized (except for those characters re-initialized by the new `basic_string`). Otherwise the `basic_stringbuf` has been created in neither input nor output mode and a zero length `basic_string` is returned.

Effects: Equivalent to: `return basic_string<charT, traits, Allocator>(view(), get_allocator());`

```
template<class SAlloc>
```

```
basic_string<charT, traits, SAlloc> str(const SAlloc& sa) const;
```

7 *Constraints:* `SAlloc` is a type that qualifies as an allocator ([container.requirements.general]).

8 *Effects:* Equivalent to: `return basic_string<charT, traits, SAlloc>(view(), sa);`

Add the following specifications for `str()`~~44~~ and `view()` *const* member function. :

```
basic_string<charT, traits, Allocator> str() &&;
```

9 *Returns:* A `basic_string<charT, traits, Allocator>` object move constructed from the `basic_stringbuf`'s underlying character sequence in `buf`. This can be achieved by first adjusting `buf` to have the same content as `view()`.

10 *Ensures:* The underlying character sequence `buf` is empty and `pbase()`, `pptr()`, `epptr()`, `eback()`, `gptr()`, `egptr()` are initialized as if by calling `init_buf_ptrs()` with an empty `buf`.

```
basic_string_view<charT, traits> view() const noexcept;
```

11 Let `sv` be `basic_string_view<charT, traits>`.

12 *Returns:* A `sv` object referring to the `basic_stringbuf`'s underlying character sequence in `buf`:

(12.1) — If `ios_base::out` is set in mode, then `sv(pbase(), high_mark-pbase())` is returned.

(12.2) — Otherwise, if `ios_base::in` is set in mode, then `sv(eback(), egptr()-eback())` is returned.

(12.3) — Otherwise, `sv()` is returned.

13 [*Note:* Using the returned `sv` object after destruction or invalidation of the character sequence underlying `*this` is undefined behavior, unless `sv.empty()` is `true`. — end note]

add setter overloads and simplify their specification trough relying on `buf` and `init_buf_ptrs()`.

```
void str(basic_string<charT, traits, Allocator>&& s);
```

14 *Effects:* Equivalent to:

```
buf = std::move(s);
init_buf_ptrs();
```

```
void str(const basic_string<charT, traits, Allocator>& s);
```

15 *Effects:* Equivalent to:

```
buf = s;
init_buf_ptrs();
```

~~Copies the content of `s` into the `basic_stringbuf` underlying character sequence and initializes the input and output sequences according to `mode`.~~

16 ~~*Ensures:* If `mode & ios_base::out` is nonzero, `pbase()` points to the first underlying character and `pptr() >= pbase() + s.size()` holds; in addition, if `mode & ios_base::ate` is nonzero, `pptr() == pbase() + s.size()` holds; otherwise `pptr() == pbase()` is true. If `mode & ios_base::in` is nonzero, `eback()` points to the first underlying character, and both `gptr() == eback()` and `egptr() == eback() + s.size()` hold.~~

```
template<class SAlloc>
void str(const basic_string<charT, traits, SAlloc>& s);
```

17 *Constraints:* `is_same_v<SAlloc, Allocator>` is false.

18 *Effects:* Equivalent to:

```
buf = s;
init_buf_ptrs();
```

6.3 28.8.3 Adjust synopsis of `basic_istream` [`istream`]

Provide constructor overloads taking an `Allocator` argument and also those that allow a string with a different allocator type.

```
// [istream.cons], constructors:
basic_istream() : basic_istream(ios_base::in) {}
explicit basic_istream(ios_base::openmode which);
explicit basic_istream(
    const basic_string<charT, traits, Allocator>& str,
    ios_base::openmode which = ios_base::in);

basic_istream(
    ios_base::openmode which,
    const Allocator& a);
explicit basic_istream(
    basic_string<charT, traits, Allocator>&& s,
    ios_base::openmode which = ios_base::in);

template <class SAlloc>
basic_istream(
    const basic_string<charT, traits, SAlloc>& s,
    const Allocator& a) : basic_istream(s, ios_base::in, a) {}
template <class SAlloc>
basic_istream(
```

```

    const basic_string<charT, traits, SAlloc>& s,
    ios_base::openmode which,
    const Allocator& a);
template <class SAlloc>
explicit basic_istreamstream(
    const basic_string<charT, traits, SAlloc>& s,
    ios_base::openmode which = ios_base::in);

basic_istreamstream(const basic_istreamstream& rhs) = delete;
basic_istreamstream(basic_istreamstream&& rhs);

// [istreamstream.assign], assign and swap
basic_istreamstream& operator=(const basic_istreamstream& rhs) = delete;
basic_istreamstream& operator=(basic_istreamstream&& rhs);
void swap(basic_istreamstream& rhs);

```

Adjust getter/setter members according to basic_stringbuf:

```

// [istreamstream.members], members:
basic_stringbuf<charT, traits, Allocator>* rdbuf() const;

basic_string<charT, traits, Allocator> str() const &;

template<class SAlloc>
basic_string<charT, traits, SAlloc> str(const SAlloc& sa) const;
basic_string<charT, traits, Allocator> str() &&;
basic_string_view<charT, traits> view() const noexcept;

void str(const basic_string<charT, traits, Allocator>& s);

template<class SAlloc>
void str(const basic_string<charT, traits, SAlloc>& s);
void str(basic_string<charT, traits, Allocator>&& s);

```

6.3.1 28.8.3.1 basic_istreamstream constructors [istreamstream.cons]

Adjust the constructor specifications analog to basic_stringbuf. deliberately do not provide the special move constructor taking an allocator. Drive-by editorial fix to include Allocator template argument.

```
explicit basic_istreamstream(ios_base::openmode which);
```

- 1 *Effects:* ~~Constructs an object of class basic_istreamstream<charT, traits>, initializing~~ Initializes the base class with basic_istream<charT, traits>(addressof(sb)) ([istream]) and ~~initializing~~ sb with basic_stringbuf<charT, traits, Allocator>(which | ios_base::in) ([stringbuf.cons]).

```
explicit basic_istreamstream(
    const basic_string<charT, traits, Allocator>& str,
    ios_base::openmode which = ios_base::in);
```

- 2 *Effects:* ~~Constructs an object of class basic_istreamstream<charT, traits>, initializing~~ Initializes the base class with basic_istream<charT, traits>(addressof(sb)) ([istream]) and ~~initializing~~ sb with basic_stringbuf<charT, traits, Allocator>(str, which | ios_base::in) ([stringbuf.cons]).

```

basic_istream(
    ios_base::openmode which,
    const Allocator& a);
3     Effects: Initializes the base class with basic_istream<charT, traits>(addressof(sb)) ([istream])
    and sb with basic_stringbuf<charT, traits, Allocator>(which | ios_base::in, a) ([string-
    buf.cons]).

explicit basic_istream(
    basic_string<charT, traits, Allocator>&& s,
    ios_base::openmode which = ios_base::in);
4     Effects: Initializes the base class with basic_istream<charT, traits>(addressof(sb)) ([istream])
    and sb with basic_stringbuf<charT, traits, Allocator>(std::move(s), which | ios_-
    base::in) ([stringbuf.cons]).

template<class SAlloc>
basic_istream(
    const basic_string<charT, traits, SAlloc>& s,
    ios_base::openmode which,
    const Allocator& a);
5     Effects: Initializes the base class with basic_istream<charT, traits>(addressof(sb)) ([istream])
    and sb with basic_stringbuf<charT, traits, Allocator>(s, which | ios_base::in, a) ([string-
    buf.cons]).

template<class SAlloc>
explicit basic_istream(
    const basic_string<charT, traits, SAlloc>& s,
    ios_base::openmode which = ios_base::in);
6     Effects: Initializes the base class with basic_istream<charT, traits>(addressof(sb)) ([istream])
    and sb with basic_stringbuf<charT, traits, Allocator>(s, which | ios_base::in) ([string-
    buf.cons]).

```

6.3.2 28.8.3.2 Assignment and swap [istream.assign]

change the definition of the `swap` member to incorporate the new requirement implicitly given by `basic_stringbuf::swap`:

```

void swap(basic_istream& rhs);
1     Effects: Exchanges the state of *this and rhs by calling Equivalent to:
    basic_istream<charT, traits>::swap(rhs); and
    sb.swap(rhs.sb);

```

6.3.3 28.8.3.3 Member functions [istream.members]

Extend `str()` overloads according to `basic_stringbuf` and add `view()`:

```

basic_string<charT, traits, Allocator> str() const &;
1     Returns: Effects: Equivalent to: return rdbuf()->str();

```



```

template<class SAlloc>
basic_string<charT,traits,SAlloc> str(const SAlloc& sa) const;
2      Effects: Equivalent to: return rdbuf()->str(sa);

basic_string<charT,traits,Allocator> str() &&;
3      Effects: Equivalent to: return std::move(*rdbuf()).str();

basic_string_view<charT, traits> view() const noexcept;
4      Effects: Equivalent to: return rdbuf()->view();

void str(const basic_string<charT, traits, Allocator>& s);
5      Effects: CallsEquivalent to: rdbuf()->str(s).

template<class SAlloc>
void str(const basic_string<charT, traits, SAlloc>& s);
6      Effects: Equivalent to: rdbuf()->str(s).

void str(basic_string<charT, traits, Allocator>&& s);
7      Effects: Equivalent to: rdbuf()->str(std::move(s)).

```

6.4 28.8.4 Adjust synopsis of basic_ostringstream [ostringstream]

Provide constructor overloads taking an Allocator argument and also those that allow a string with a different allocator type.

```

// [ostringstream.cons], constructors:
basic_ostringstream() : basic_ostringstream(ios_base::in) {}
explicit basic_ostringstream(ios_base::openmode which);
explicit basic_ostringstream(
    const basic_string<charT, traits, Allocator>& str,
    ios_base::openmode which = ios_base::out);

basic_ostringstream(
    ios_base::openmode which,
    const Allocator& a);
explicit basic_ostringstream(
    basic_string<charT, traits, Allocator>&& s,
    ios_base::openmode which = ios_base::out);

template <class SAlloc>
basic_ostringstream(
    const basic_string<charT, traits, SAlloc>& s,
    const Allocator& a) : basic_ostringstream(s, ios_base::out, a) {}
template <class SAlloc>
basic_ostringstream(
    const basic_string<charT, traits, SAlloc>& s,
    ios_base::openmode which,
    const Allocator& a);
template <class SAlloc>
explicit basic_ostringstream(

```

```

    const basic_string<charT, traits, SAlloc>& s,
    ios_base::openmode which = ios_base::out);

basic_ostringstream(const basic_ostringstream& rhs) = delete;
basic_ostringstream(basic_ostringstream&& rhs);

// [ostream.assign], assign and swap
basic_ostringstream& operator=(const basic_ostringstream& rhs) = delete;
basic_ostringstream& operator=(basic_ostringstream&& rhs);
void swap(basic_ostringstream& rhs);

```

Adjust getter/setter members according to basic_stringbuf:

```

// [ostream.members], members:
basic_stringbuf<charT, traits, Allocator>* rdbuf() const;

basic_string<charT, traits, Allocator> str() const &;

template<class SAlloc>
basic_string<charT, traits, SAlloc> str(const SAlloc& sa) const;
basic_string<charT, traits, Allocator> str() &&;
basic_string_view<charT, traits> view() const noexcept;

void str(const basic_string<charT, traits, Allocator>& s);

template<class SAlloc>
void str(const basic_string<charT, traits, SAlloc>& s);
void str(basic_string<charT, traits, Allocator>&& s);

```

6.4.1 28.8.4.1 basic_ostringstream constructors [ostream.cons]

Adjust the constructor specifications analog to basic_stringbuf. deliberately do not provide the special move constructor taking an allocator. Drive-by editorial fix to include Allocator template argument.

```
explicit basic_ostringstream(ios_base::openmode which);
```

- 1 *Effects:* ~~Constructs an object of class `basic_ostringstream<charT, traits>`, initializing~~ Initializes the base class with `basic_ostream<charT, traits>(addressof(sb))` ([ostream]) and ~~initializing~~ initializes `sb` with `basic_stringbuf<charT, traits, Allocator>(which | ios_base::out)` ([stringbuf.cons]).

```
explicit basic_ostringstream(
    const basic_string<charT, traits, Allocator>& str,
    ios_base::openmode which = ios_base::out);
```

- 2 *Effects:* ~~Constructs an object of class `basic_ostringstream<charT, traits>`, initializing~~ Initializes the base class with `basic_ostream<charT, traits>(addressof(sb))` ([ostream]) and ~~initializing~~ initializes `sb` with `basic_stringbuf<charT, traits, Allocator>(str, which | ios_base::out)` ([stringbuf.cons]).

```
basic_ostringstream(
    ios_base::openmode which,
    const Allocator& a);
```

- 3 *Effects:* Initializes the base class with `basic_ostream<charT, traits>(addressof(sb))` ([os-

tream]) and sb with basic_stringbuf<charT, traits, Allocator>(which | ios_base::out, a) ([stringbuf.cons]).

```
explicit basic_ostringstream(
    basic_string<charT, traits, Allocator>&& s,
    ios_base::openmode which = ios_base::out);
```

- 4 *Effects:* Initializes the base class with basic_ostream<charT, traits>(addressof(sb))([ostream]) and sb with basic_stringbuf<charT, traits, Allocator>(std::move(s), which | ios_base::out) ([stringbuf.cons]).

```
template<class SAlloc>
basic_ostringstream(
    const basic_string<charT, traits, SAlloc>& s,
    ios_base::openmode which,
    const Allocator& a);
```

- 5 *Effects:* Initializes the base class with basic_ostream<charT, traits>(addressof(sb))([ostream]) and sb with basic_stringbuf<charT, traits, Allocator>(s, which | ios_base::out, a) ([stringbuf.cons]).

```
template<class SAlloc>
explicit basic_ostringstream(
    const basic_string<charT, traits, SAlloc>& s,
    ios_base::openmode which = ios_base::out);
```

- 6 *Constraints:* is_same_v<SAlloc, Allocator> is false.

- 7 *Effects:* Initializes the base class with basic_ostream<charT, traits>(addressof(sb))([ostream]) and sb with basic_stringbuf<charT, traits, Allocator>(s, which | ios_base::out) ([stringbuf.cons]).

6.4.2 28.8.4.2 Assignment and swap [ostreamstream.assign]

change the definition of the swap member to incorporate the new requirement implicitly given by basic_stringbuf::swap:

```
void swap(basic_ostringstream& rhs);
```

- 1 *Effects:* ~~Exchanges the state of *this and rhs by calling~~ Equivalent to:
 basic_ostream<charT, traits>::swap(rhs); ~~and~~
 sb.swap(rhs.sb); ~~;~~

6.4.3 28.8.4.3 Member functions [ostreamstream.members]

Extend str() overloads according to basic_stringbuf and add view():

```
basic_string<charT, traits, Allocator> str() const &;
```

- 1 ~~Returns:-~~ Effects: Equivalent to: return rdbuf()->str(); ~~;~~

```
template<class SAlloc>
basic_string<charT, traits, SAlloc> str(const SAlloc& sa) const;
```

- 2 *Effects:* Equivalent to: return rdbuf()->str(sa);

```

basic_string<charT,traits,Allocator> str() &&;
3     Effects: Equivalent to: return std::move(*rdbuf()).str();

basic_string_view<charT, traits> view() const noexcept;
4     Effects: Equivalent to: return rdbuf()->view();

void str(const basic_string<charT, traits, Allocator>& s);
5     Effects: CallsEquivalent to: rdbuf()->str(s).

template<class SAlloc>
void str(const basic_string<charT, traits, SAlloc>& s);
6     Effects: Equivalent to: rdbuf()->str(s).

void str(basic_string<charT, traits, Allocator>&& s);
7     Effects: Equivalent to: rdbuf()->str(std::move(s)).

```

6.5 28.8.5 Adjust synopsis of basic_stringstream [stringstream]

Provide constructor overloads taking an Allocator argument and also those that allow a string with a different allocator type.

```

// [stringstream.cons], constructors:
basic_stringstream() : basic_stringstream(ios_base::out | ios_base::in) {}
explicit basic_stringstream(ios_base::openmode which);
explicit basic_stringstream(
    const basic_string<charT, traits, Allocator>& str,
    ios_base::openmode which = ios_base::out | ios_base::in);

basic_stringstream(
    ios_base::openmode which,
    const Allocator& a);
explicit basic_stringstream(
    basic_string<charT, traits, Allocator>&& s,
    ios_base::openmode which = ios_base::out | ios_base::in);

template <class SAlloc>
basic_stringstream(
    const basic_string<charT, traits, SAlloc>& s,
    const Allocator& a) : basic_stringstream(s, ios_base::out | ios_base::in, a) {}
template <class SAlloc>
basic_stringstream(
    const basic_string<charT, traits, SAlloc>& s,
    ios_base::openmode which,
    const Allocator& a);
template <class SAlloc>
explicit basic_stringstream(
    const basic_string<charT, traits, SAlloc>& s,
    ios_base::openmode which = ios_base::out | ios_base::in);

basic_stringstream(const basic_stringstream& rhs) = delete;
basic_stringstream(basic_stringstream&& rhs);

```

```
// [stringstream.assign], assign and swap
basic_stringstream& operator=(const basic_stringstream& rhs) = delete;
basic_stringstream& operator=(basic_stringstream&& rhs);
void swap(basic_stringstream& rhs);
```

Adjust getter/setter members according to `basic_stringbuf`:

```
// [stringstream.members], members:
basic_stringbuf<charT, traits, Allocator>* rdbuf() const;

basic_string<charT, traits, Allocator> str() const &;

template<class SAlloc>
basic_string<charT, traits, SAlloc> str(const SAlloc& sa) const;
basic_string<charT, traits, Allocator> str() &&;
basic_string_view<charT, traits> view() const noexcept;

void str(const basic_string<charT, traits, Allocator>& s);

template<class SAlloc>
void str(const basic_string<charT, traits, SAlloc>& s);
void str(basic_string<charT, traits, Allocator>&& s);
```

6.5.1 28.8.5.1 `basic_stringstream` constructors [stringstream.cons]

Adjust the constructor specifications analog to `basic_stringbuf`. deliberately do not provide the special move constructor taking an allocator. Drive-by editorial fix to include `Allocator` template argument.

```
explicit basic_stringstream(ios_base::openmode which);
```

- 1 *Effects:* ~~Constructs an object of class `basic_stringstream<charT, traits>`, initializing~~ Initializes the base class with `basic_iostream<charT, traits>(addressof(sb))` ([iostream.cons]) and initializing `sb` with `basic_stringbuf<charT, traits, Allocator>(which)` ([stringbuf.cons]).

```
explicit basic_stringstream(
    const basic_string<charT, traits, Allocator>& str,
    ios_base::openmode which = ios_base::out | ios_base::in);
```

- 2 *Effects:* ~~Constructs an object of class `basic_stringstream<charT, traits>`, initializing~~ Initializes the base class with `basic_iostream<charT, traits>(addressof(sb))` ([iostream.cons]) and ~~initializing~~ `sb` with `basic_stringbuf<charT, traits, Allocator>(str, which)` ([stringbuf.cons]).

```
basic_stringstream(
    ios_base::openmode which,
    const Allocator& a);
```

- 3 *Effects:* Initializes the base class with `basic_iostream<charT, traits>(addressof(sb))` ([iostream.cons]) and `sb` with `basic_stringbuf<charT, traits, Allocator>(which, a)` ([stringbuf.cons]).

```
explicit basic_stringstream(
    basic_string<charT, traits, Allocator>&& s,
    ios_base::openmode which = ios_base::out | ios_base::in);
```

- 4 *Effects:* Initializes the base class with `basic_iostream<charT, traits>(addressof(sb))` ([iostream.cons]) and `sb` with `basic_stringbuf<charT, traits, Allocator>(std::move(s), which)` ([stringbuf.cons]).

```

template<class SAlloc>
basic_stringstream(
    const basic_string<charT, traits, SAlloc>& s,
    ios_base::openmode which,
    const Allocator& a);
5      Effects: Initializes the base class with basic_iostream<charT, traits>(addressof(sb)) ([iostream.cons])
    and sb with basic_stringbuf<charT, traits, Allocator>(s, which, a) ([stringbuf.cons]).

template<class SAlloc>
explicit basic_stringstream(
    const basic_string<charT, traits, SAlloc>& s,
    ios_base::openmode which = ios_base::out | ios_base::in);
6      Constraints: is_same_v<SAlloc, Allocator> is false.
7      Effects: Initializes the base class with basic_iostream<charT, traits>(addressof(sb)) ([iostream.cons])
    and sb with basic_stringbuf<charT, traits, Allocator>(s, which) ([stringbuf.cons]).

```

6.5.2 28.8.5.2 Assignment and swap [stringstream.assign]

change the definition of the `swap` member to incorporate the new requirement implicitly given by `basic_stringbuf::swap`:

```

void swap(basic_stringstream& rhs);
1      Effects: Exchanges the state of *this and rhs by calling Equivalent to:
    basic_iostream<charT, traits>::swap(rhs); and
    sb.swap(rhs.sb); ;

```

6.5.3 28.8.5.3 Member functions [stringstream.members]

Extend `str()` overloads according to `basic_stringbuf` and add `view()`:

```

basic_string<charT, traits, Allocator> str() const &;
1      Returns: Effects: Equivalent to: return rdbuf()->str();

template<class SAlloc>
basic_string<charT, traits, SAlloc> str(const SAlloc& sa) const;
2      Effects: Equivalent to: return rdbuf()->str(sa);

basic_string<charT, traits, Allocator> str() &&;
3      Effects: Equivalent to: return std::move(*rdbuf()).str();

basic_string_view<charT, traits> view() const noexcept;
4      Effects: Equivalent to: return rdbuf()->view();

void str(const basic_string<charT, traits, Allocator>& s);
5      Effects: Calls Equivalent to: rdbuf()->str(s).

template<class SAlloc>

```

```
void str(const basic_string<charT, traits, SAlloc>& s);  
6      Effects: Equivalent to: rdbuf()->str(s).  
  
void str(basic_string<charT, traits, Allocator>&& s);  
7      Effects: Equivalent to: rdbuf()->str(std::move(s)).
```

7 Appendix: Example Implementations

The given specification has been implemented within a recent version of the sstream header of gcc8. Modified version of the headers and some tests are available at https://github.com/PeterSommerlad/SC22WG21_Papers/tree/master/workspace/Test_basic_stringbuf_efficient/src.

A corresponding implementation for clang 7 is available in the vicinity of the one above at: https://github.com/PeterSommerlad/SC22WG21_Papers/tree/master/workspace/Test_clang_p0407_p0408